

壓縮機簡介

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壓縮機簡介

壓縮機功能

壓縮機的功能是在於對氣體增壓，以使氣體能輸送到某一高壓地點或較遠的地方以克服管路磨擦損失，或配合某種化學反應。

壓縮機簡介

壓縮機分類

由於氣體具有可壓縮性，會因氣體流量、進出口壓力及比速率(Specific Speed)等因素，而會選用不同的壓縮機的機型。

壓縮機的機型可分為：

- 動力式(Dynamic Type)
- 正排量式(Positive Displacement Type)



壓縮機簡介

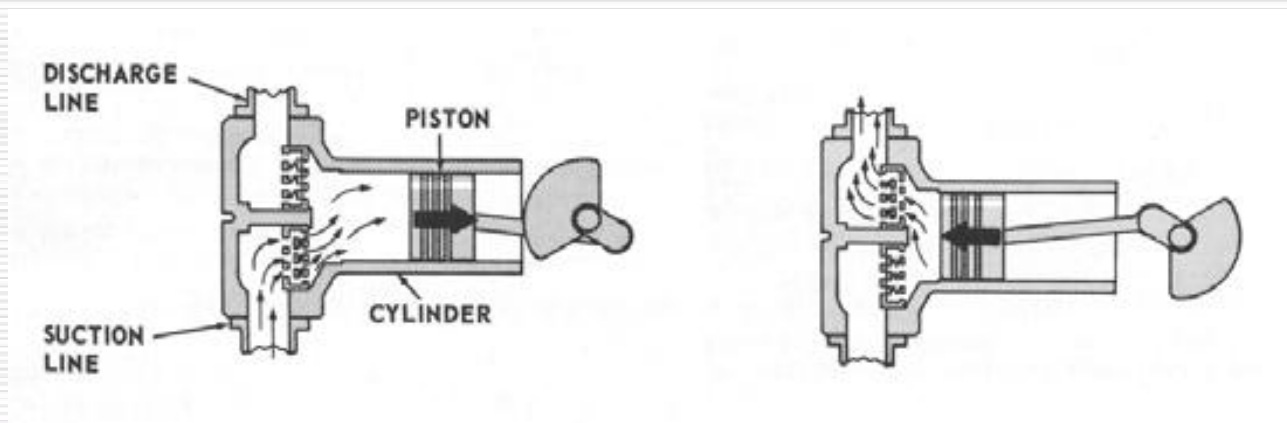
正排量壓縮機簡介

- 正排量壓縮機工作原理
- 正排量壓縮機分類
- 各型式正排量壓縮機簡介

壓縮機簡介

正排量壓縮機工作原理

藉由一個封閉空間，對其體積做增大與減小的工作，依膨脹、進氣、收縮(壓縮)、排氣的程序，對氣體進行壓縮的工作。



壓縮機簡介

正排量壓縮機分類



壓縮機簡介

正排量壓縮機類型例說-

氣冷往復式

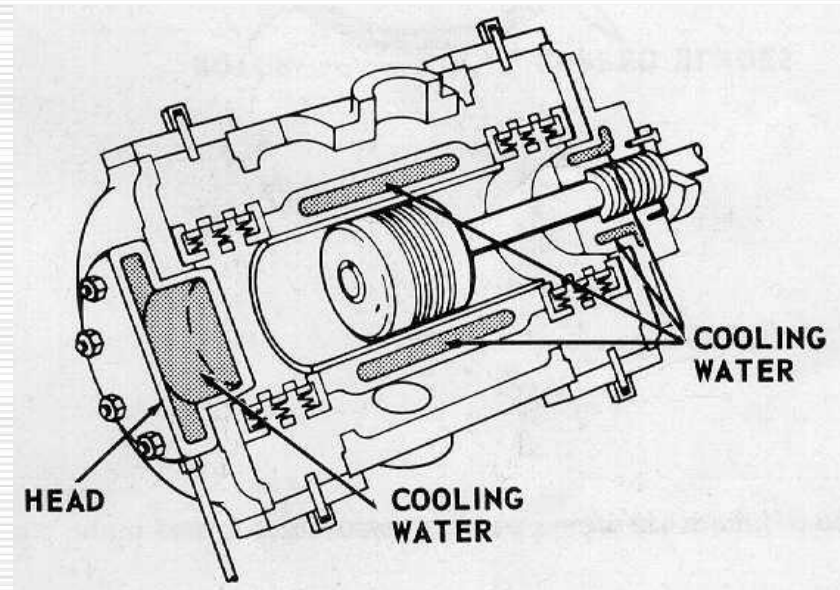
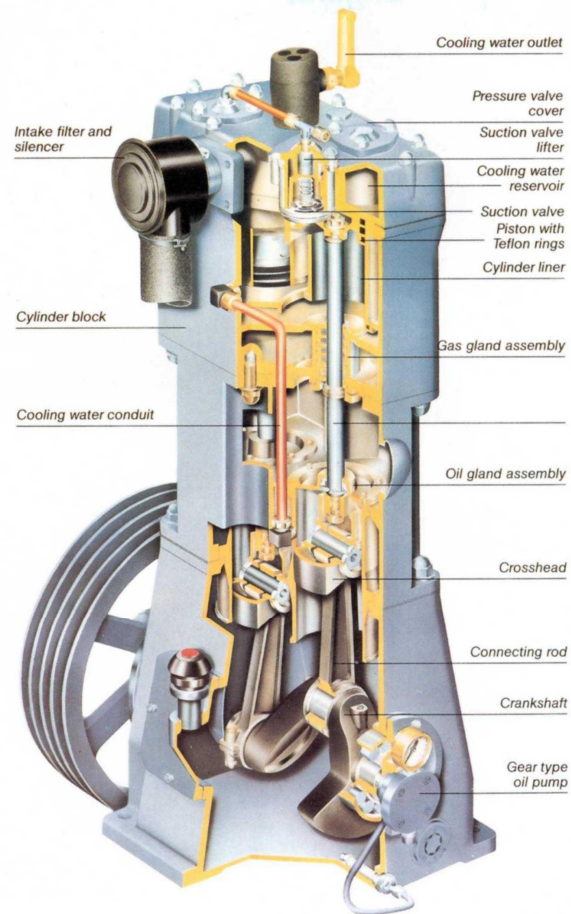


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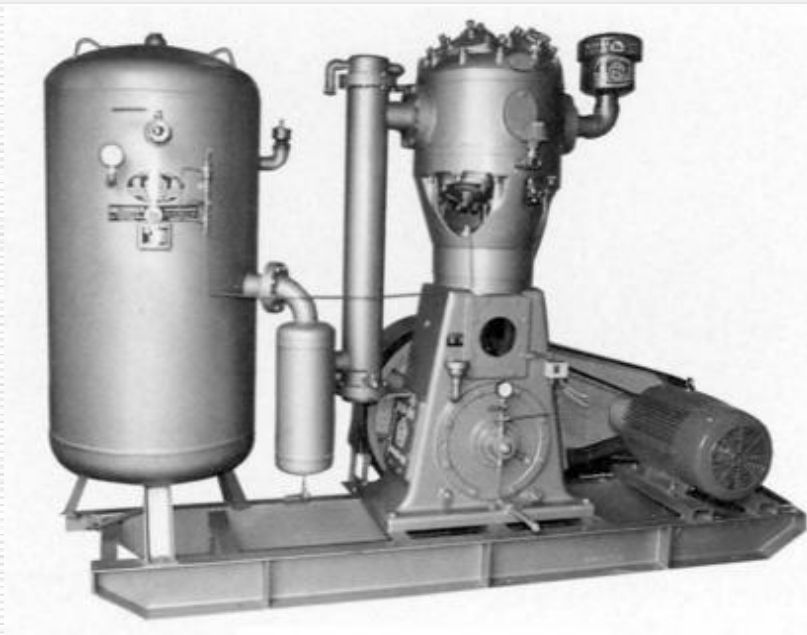
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正排量壓縮機類型 例說-水冷往復式

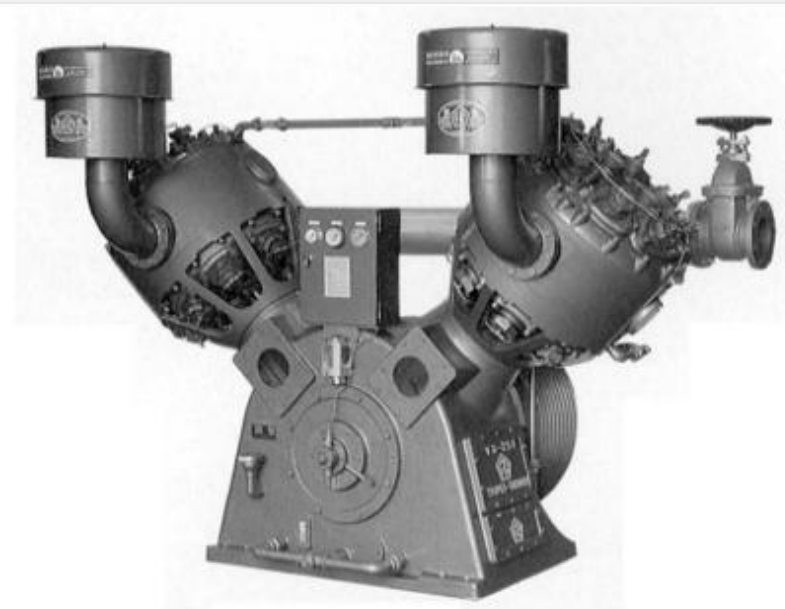


壓縮機簡介

正排量壓縮機類型例說-單級往復式



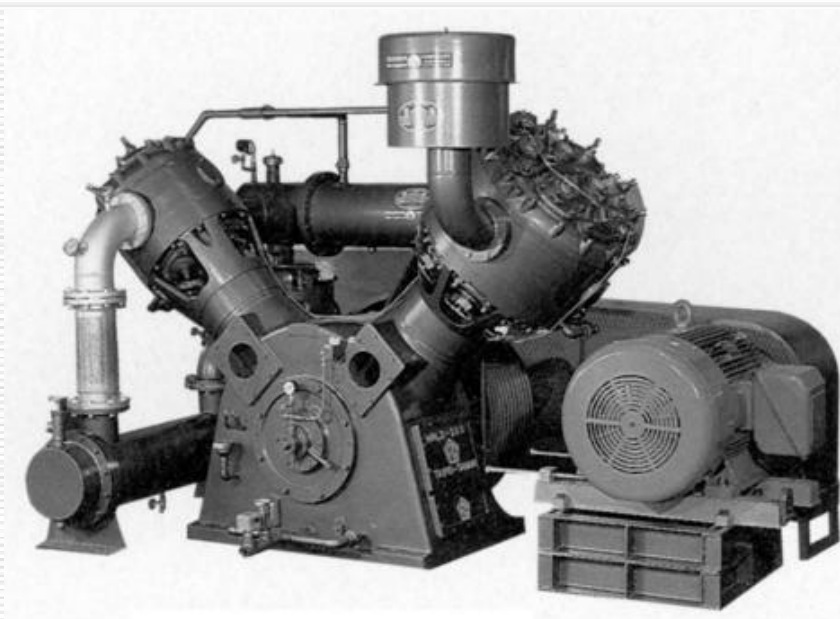
SINGLE-STAGE RECIPROCATING COMPRESSOR



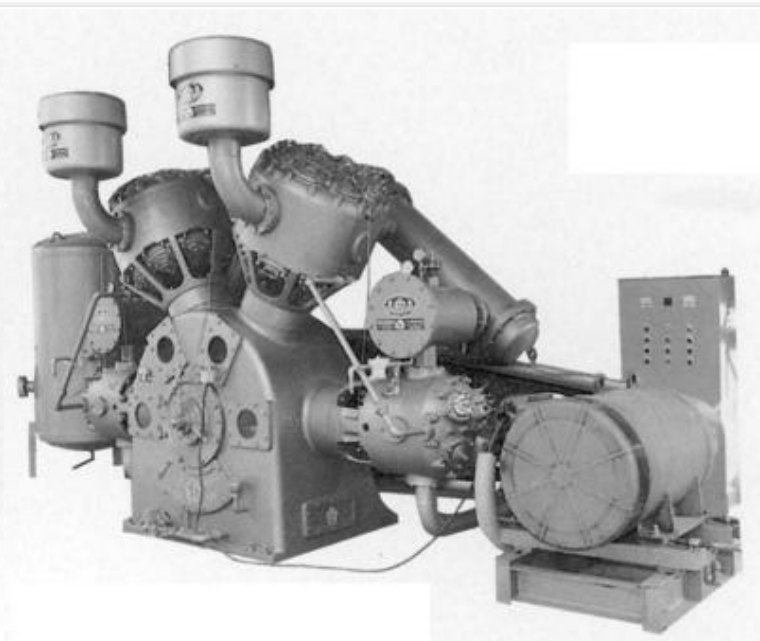
SINGLE-STAGE TWO CYLINDERS RECIPROCATING
COMPRESSOR

壓縮機簡介

正排量壓縮機類型例說-多級往復式



MUTISTAGE RECIPROCATING COMPRESSOR



MUTISTAGE RECIPROCATING COMPRESSOR

壓縮機簡介

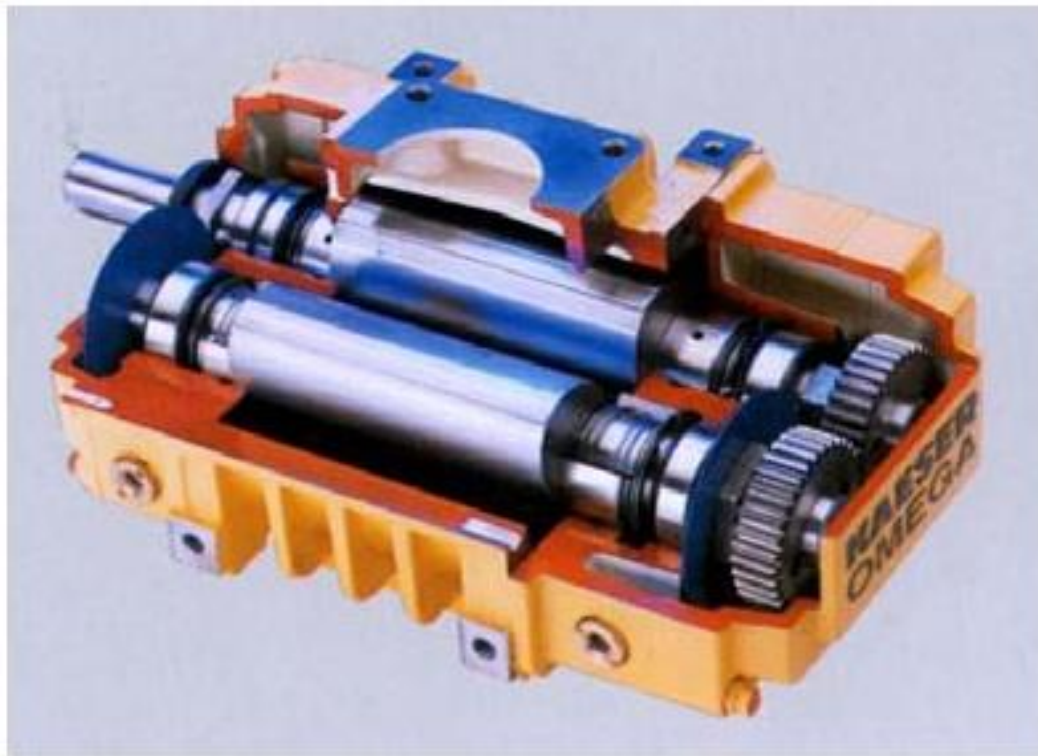
正排量壓縮機-往復式壓縮機輔助系統

- 閉迴路冷卻水系統
- 潤滑油單元

壓縮機簡介

正排量壓縮機類型例說-

瓣式 (Lobe Type)



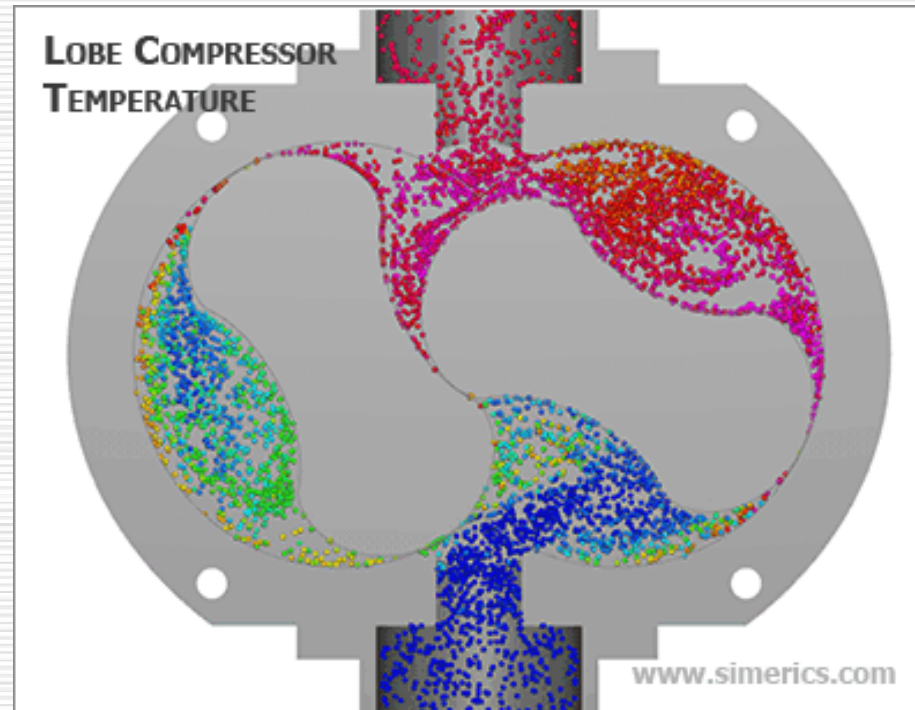
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壓縮機簡介

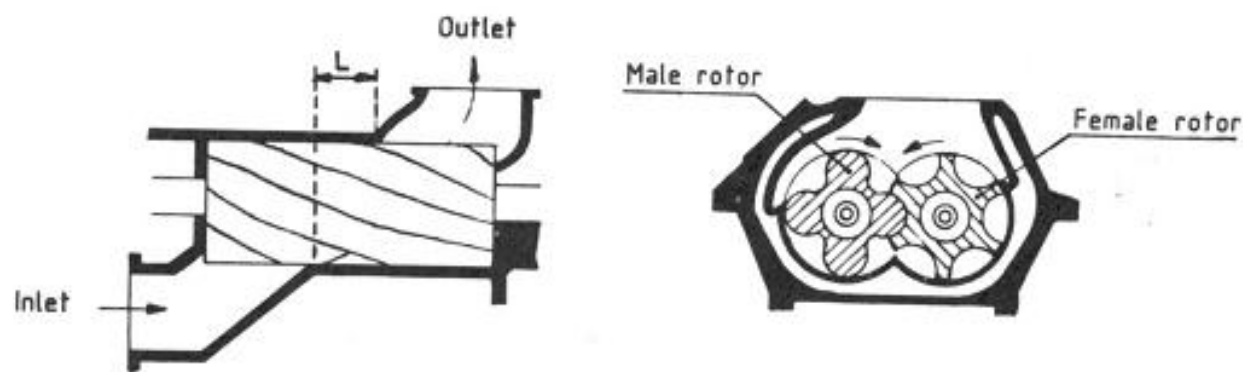
正排量壓縮機類型例說-

瓣式 (Lobe Type)



壓縮機簡介

正排量壓縮機類型例說- 螺旋式 (Screw Type)

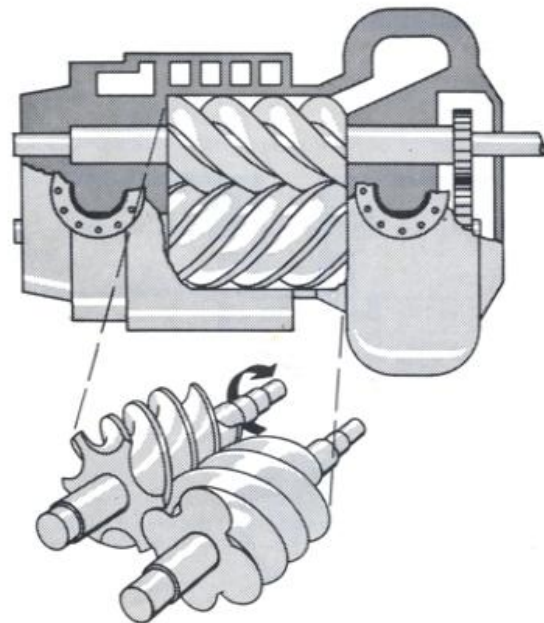


OPERATING PRINCIPLE OF THE SCREW COMPRESSOR

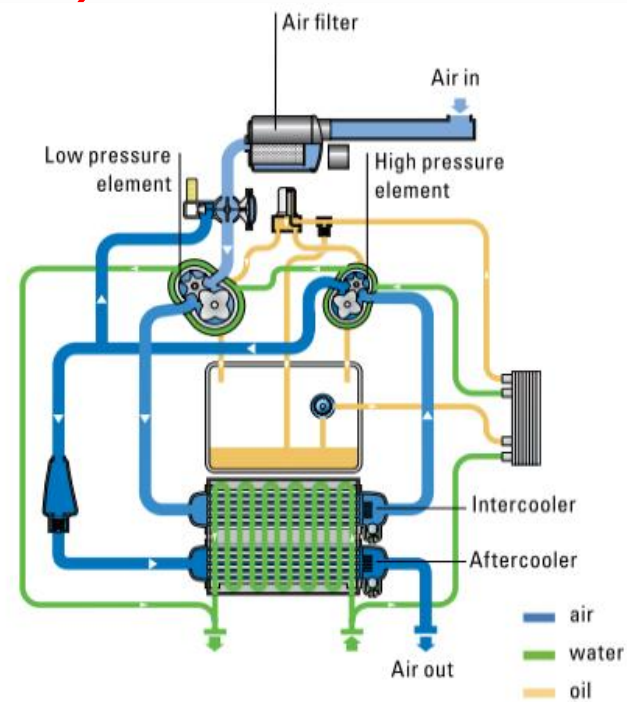
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正排量壓縮機類型例說-

螺旋式 (Screw Type; Oil Free)



DRY SCREW COMPRESSOR



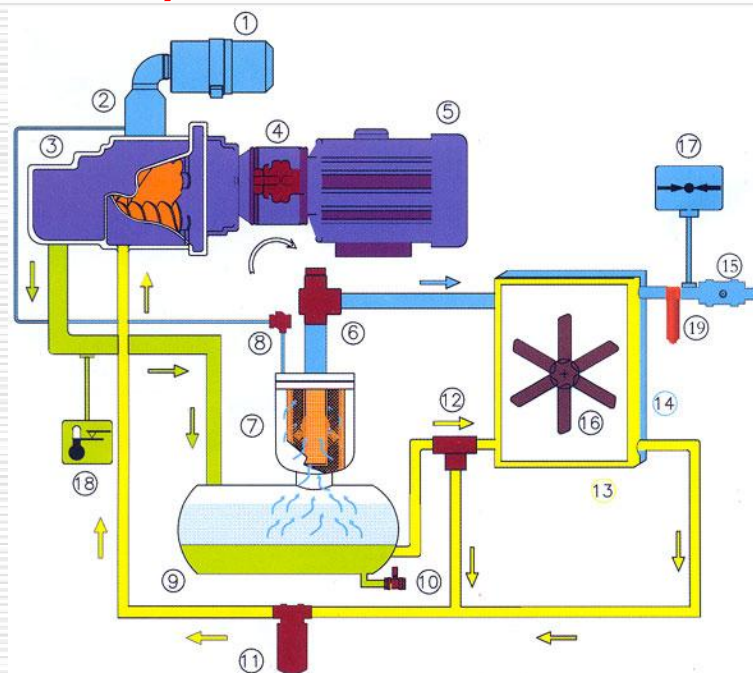
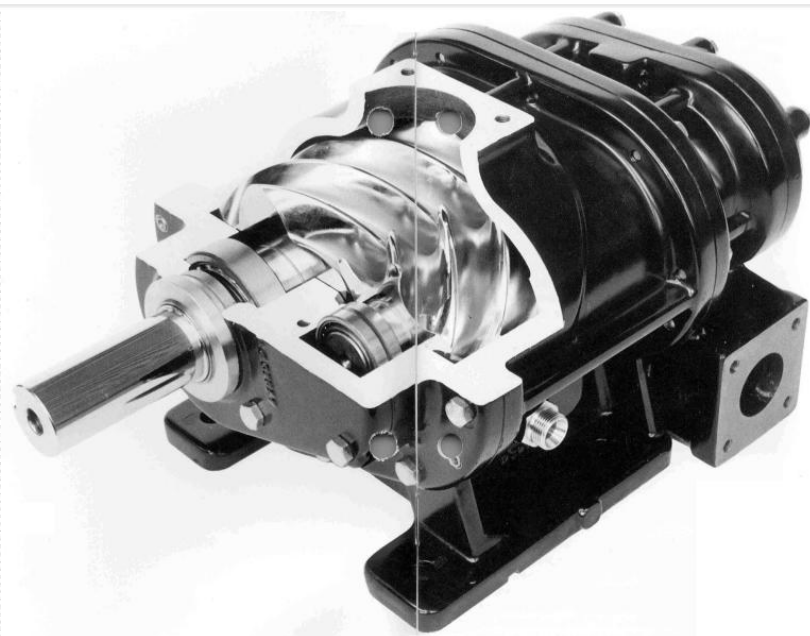
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<Reference> Atlas Copco Product Brochure for the picture on right side.



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正排量壓縮機類型例說- 螺旋式 (Screw Type; Oil Flooded)



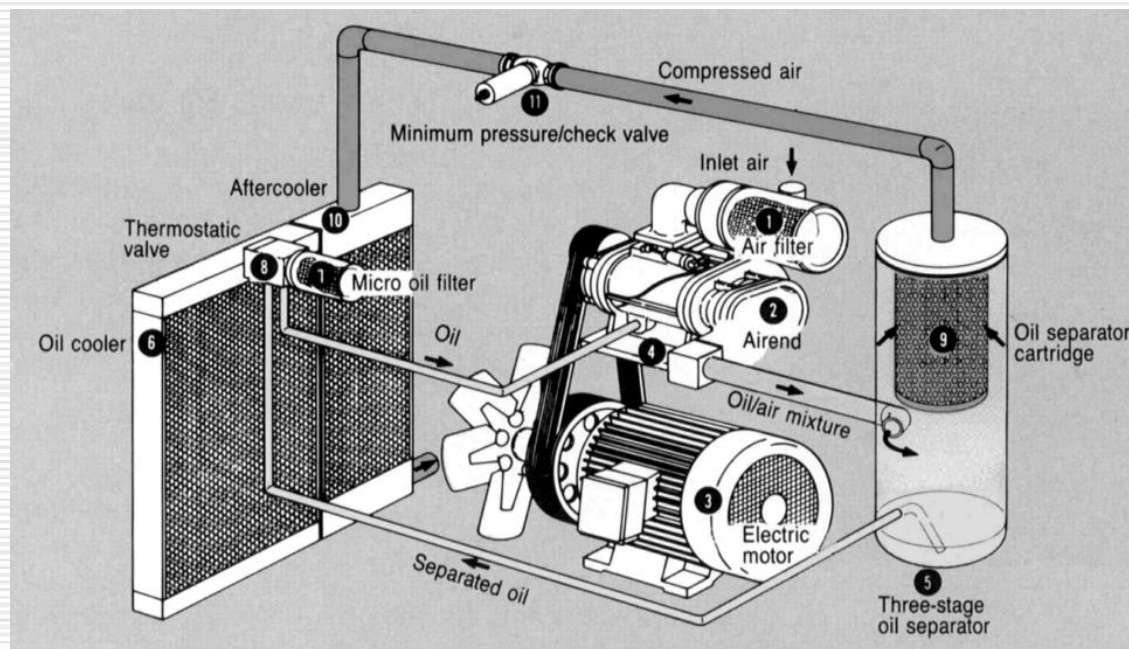
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<Reference> <http://www.poonapneumatic.com> for the picture on right side.



壓縮機簡介

正排量壓縮機類型例說- 螺旋式 (Screw Type; Oil Flooded)



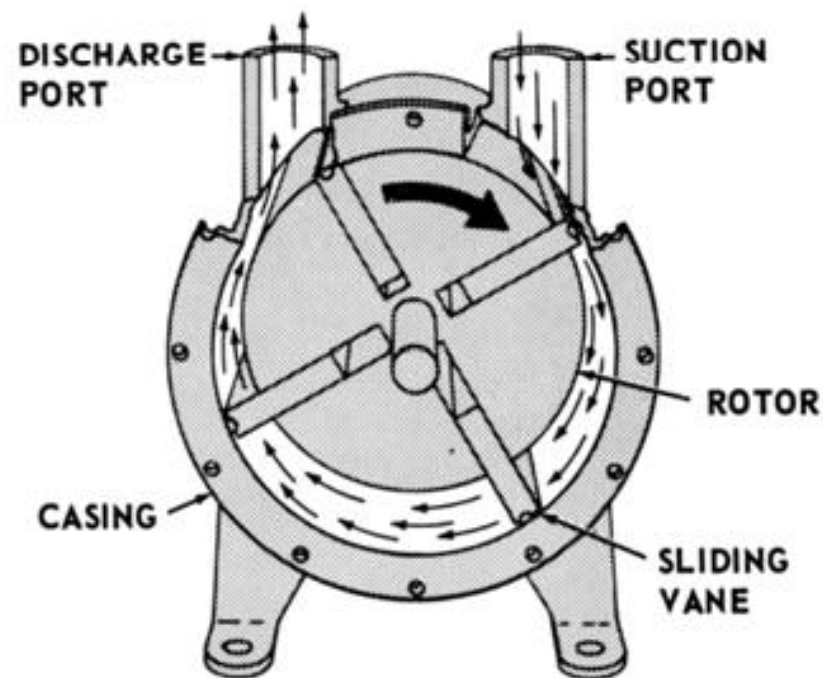
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壓縮機簡介

正排量壓縮機類型例說-

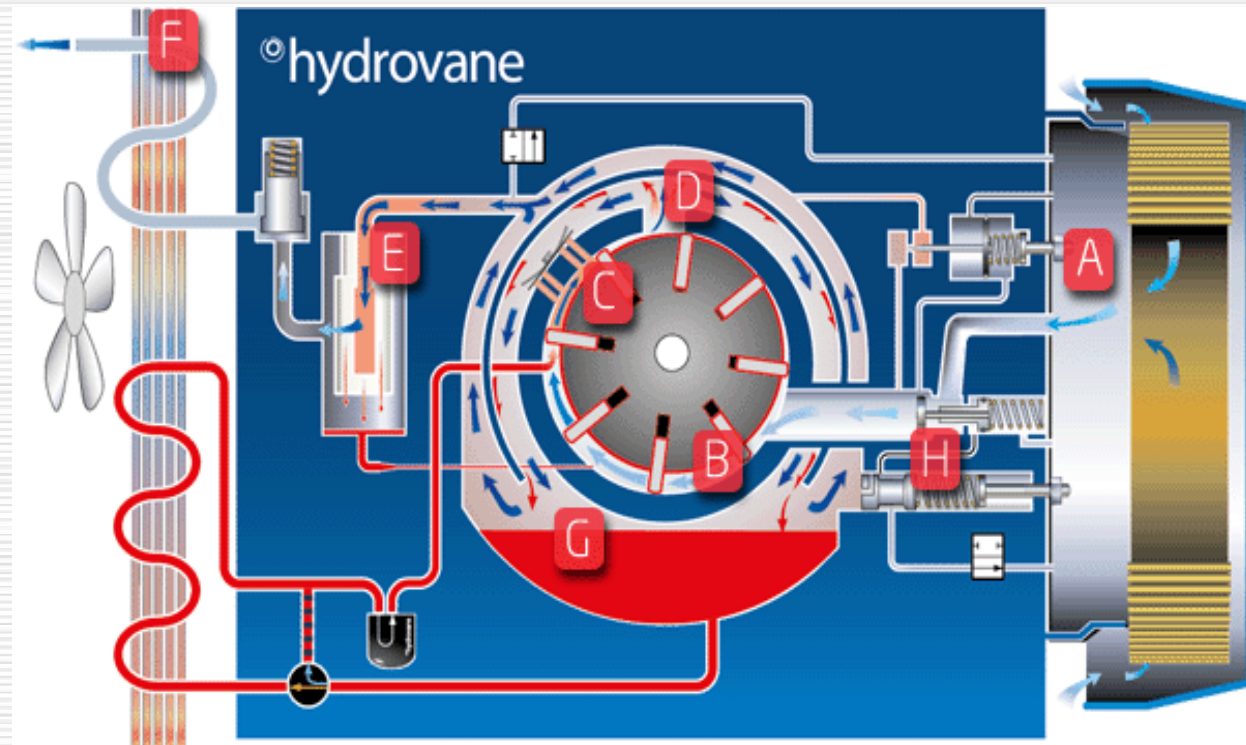
翼片式 (Vane Type)



壓縮機簡介

正排量壓縮機類型例說-

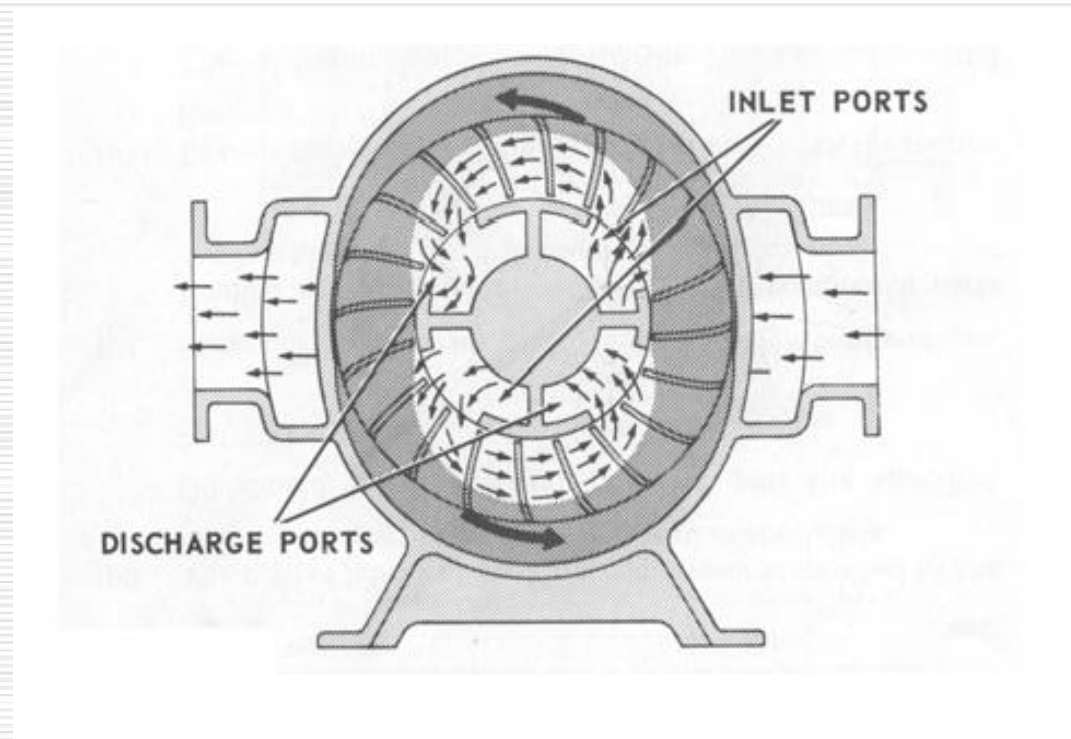
翼片式
(Vane Type)



壓縮機簡介

正排量壓縮機類型例說-

液封式
(Liquid Ring)



壓縮機簡介

正排量壓縮機類型例說- 液封式(Liquid Ring)

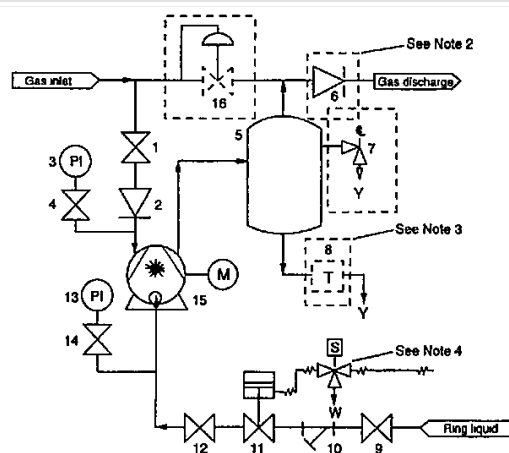


Figure A—LRVP Once Through System

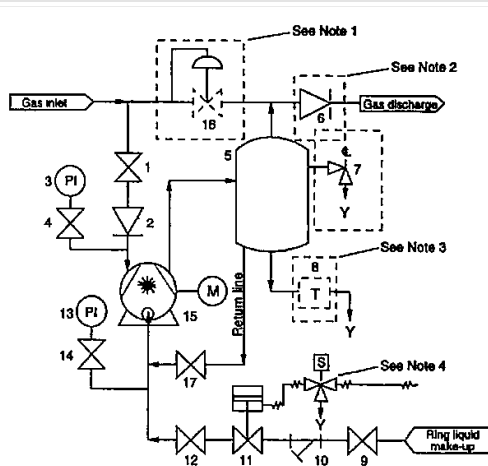


Figure B—LRVP Partial Recirculation System

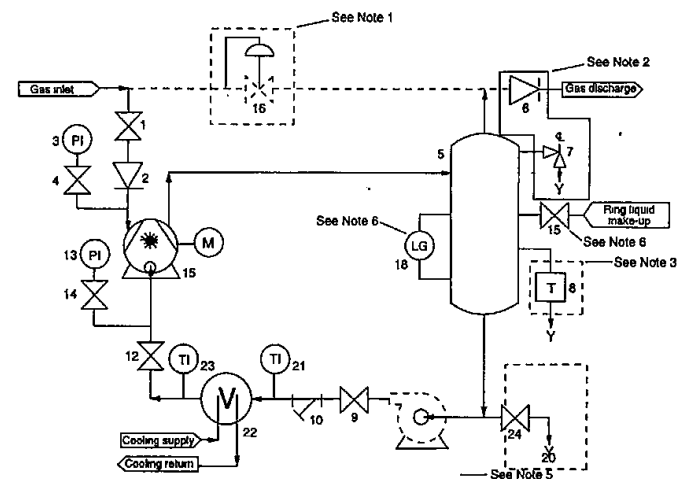


Figure C—LRVP Total Recirculation System

Figure D-1—Liquid Ring Vacuum Pump Ring Liquid Flow Schematics



壓縮機簡介

正排量壓縮機類型例說-

Figures A & B

1. Inlet isolating valve.
2. Inlet check valve.
3. Inlet pressure gauge.
4. Gauge isolation valve.
5. Gas/ring liquid separator.
6. Discharge check valve (see Note 2).
7. Pressure relief valve (see Note 2).
8. Ring liquid drain trap (see Note 3).
9. Inlet isolating valve.
10. Y Strainer.
11. Pilot operated automatic valve (see Note 4).
12. Flow regulating valve or orifice.
13. Ring liquid pressure gauge.
14. Gauge isolating valve.
15. Liquid ring pump and drive.
16. Inlet pressure control (see Note 1).
17. Partial ring liquid recirculation valve (Figure B only).

Figure C-Additions

18. Level gauge.
19. Make-up valve.
20. Optional ring liquid recirculation pump.
21. Temperature gauge.
22. Heat exchanger.
23. Temperature gauge.
24. Drain valve.

液封式 (Liquid Ring)

Notes

1. Optional inlet pressure control valve.
2. Optional pressurized discharge check valve and pressure relief valve. (Required on liquid ring compressors or vacuum pumps with positive discharge to a contained system.)
3. Optional pressurized discharge ring liquid drain system. (Required for compressors.)
4. Pilot operated isolating valve. Direct acting solenoid may be substituted at purchaser's discretion for small sizes.
5. Recirculation pump optional based on purchaser's requirements or application details.
6. Optional level controls may be specified by purchaser.



壓縮機簡介

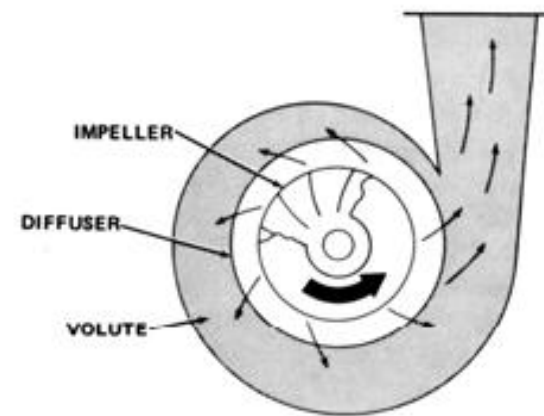
動力式壓縮機簡介

- 動力式壓縮機工作原理
- 動力式壓縮機分類
- 各型式動力式壓縮機簡介
- 輔助單元概述

壓縮機簡介

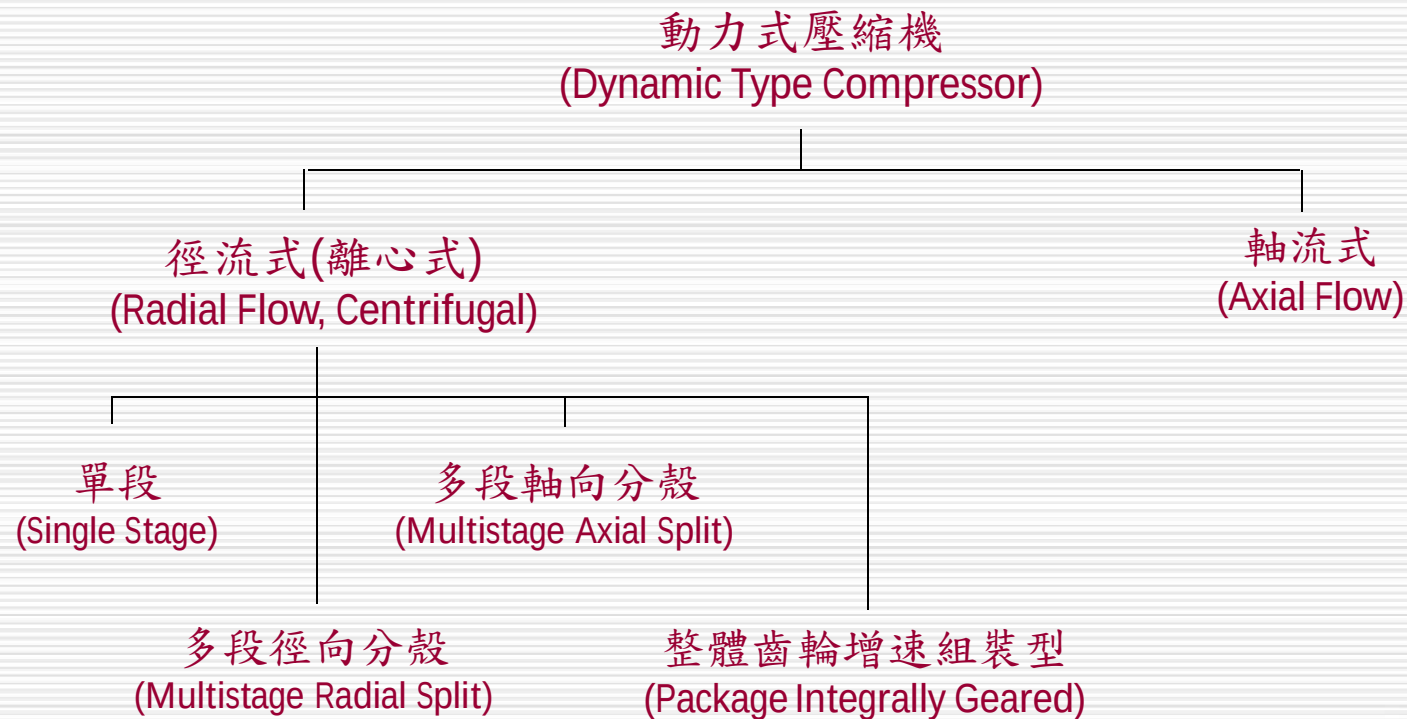
動力式壓縮機工作原理

藉迴轉葉輪對氣體做功，
使氣體的速度增快，再
依熱力學擴散管的原理，
將氣體減速而增壓。



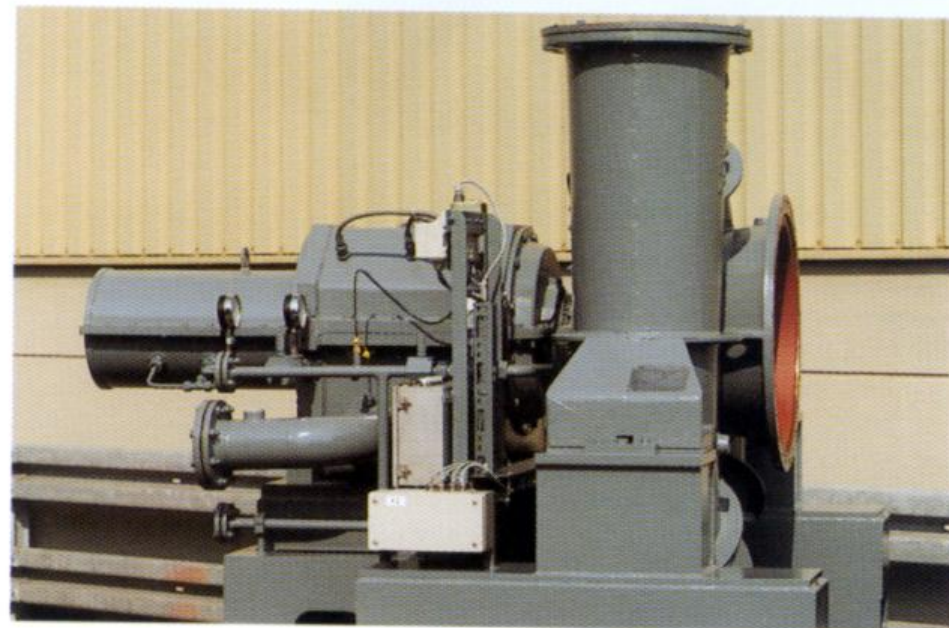
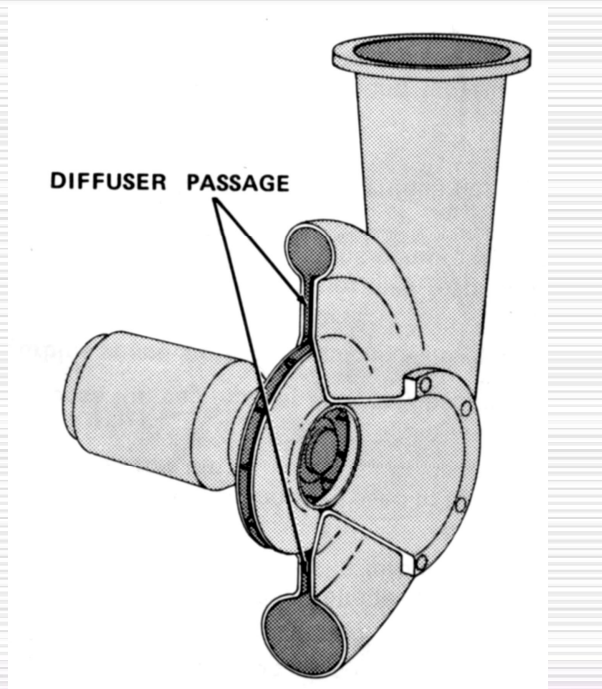
壓縮機簡介

動力式壓縮機工作分類



壓縮機簡介

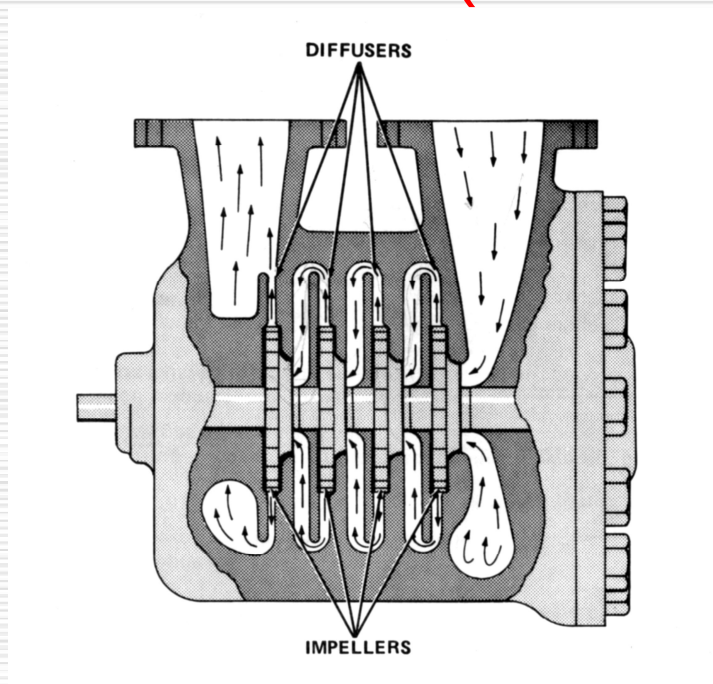
各型式動力式壓縮機簡介- 單段徑流式 (Single Stage Radial Flow)



壓縮機簡介

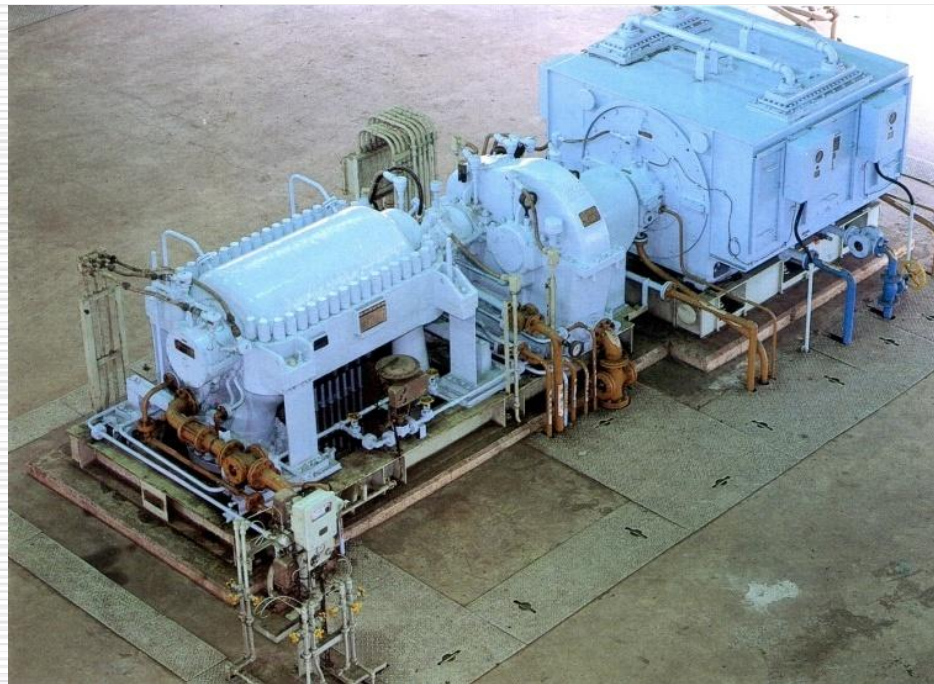
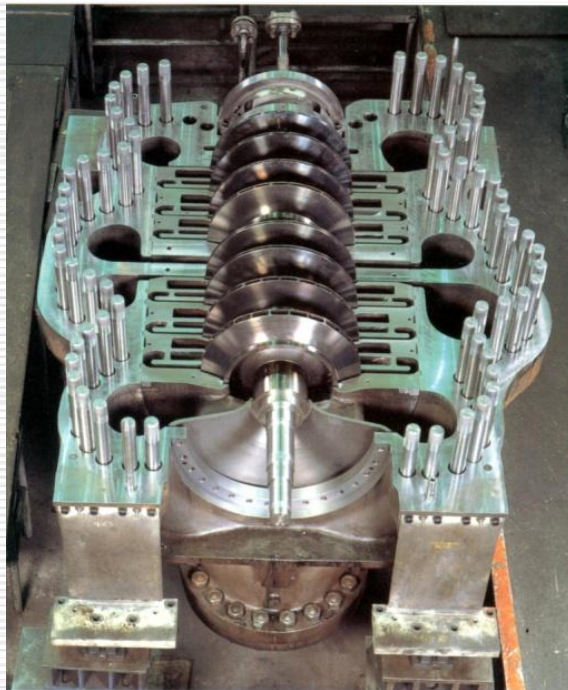
各型式動力式壓縮機簡介-

多段徑向分殼 (Multistage Radial Split)



壓縮機簡介

各型式動力式壓縮機簡介- 多段軸向分殼 (Multistage Axial Split)



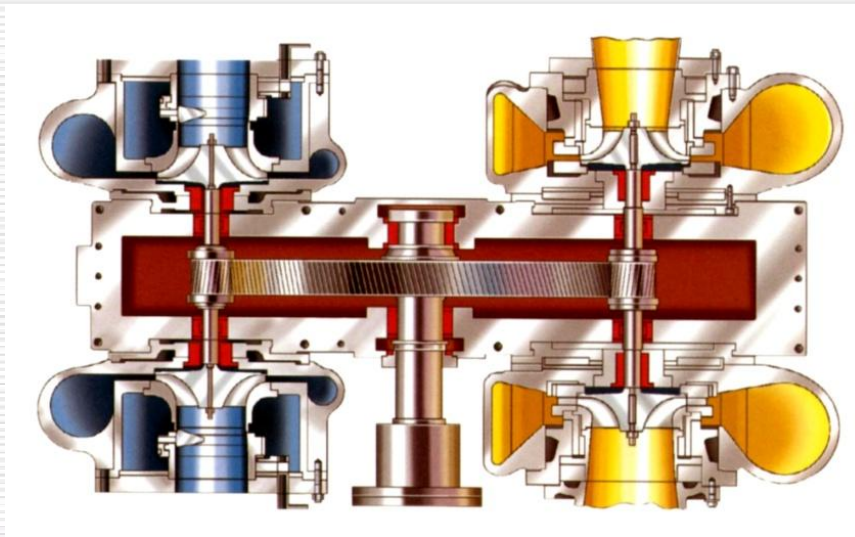
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壓縮機簡介

各型式動力式壓縮機簡介-

整體齒輪增速組裝型
(Packaged, Integrally Geared)



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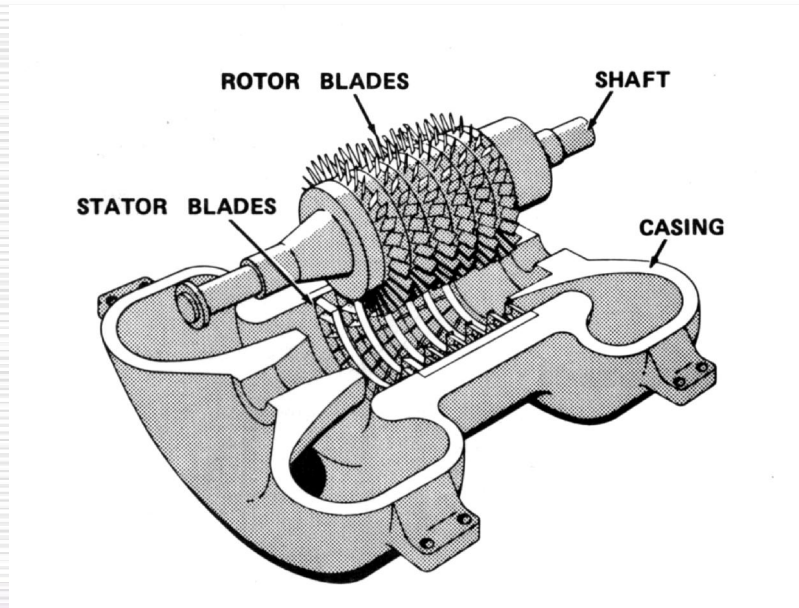
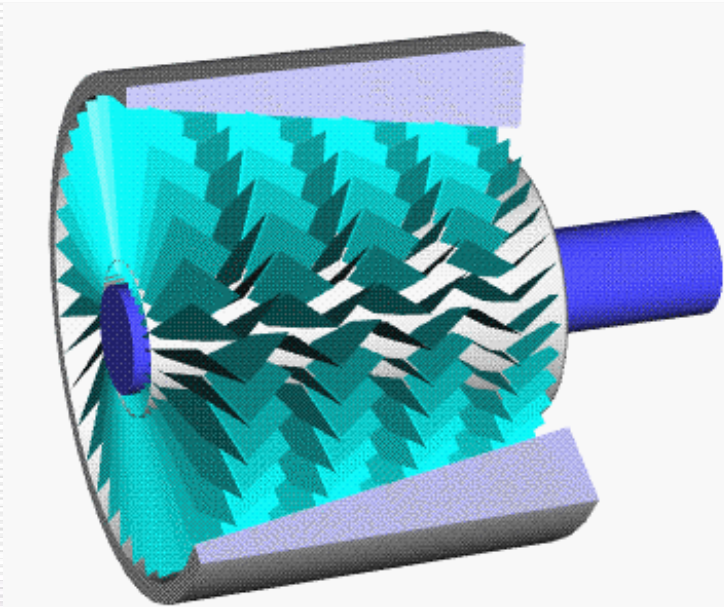


壓縮機簡介

各型式動力式壓縮機簡介-

軸流式, 固定攻角靜葉

(Axial Type, Fixed Stator Vanes)



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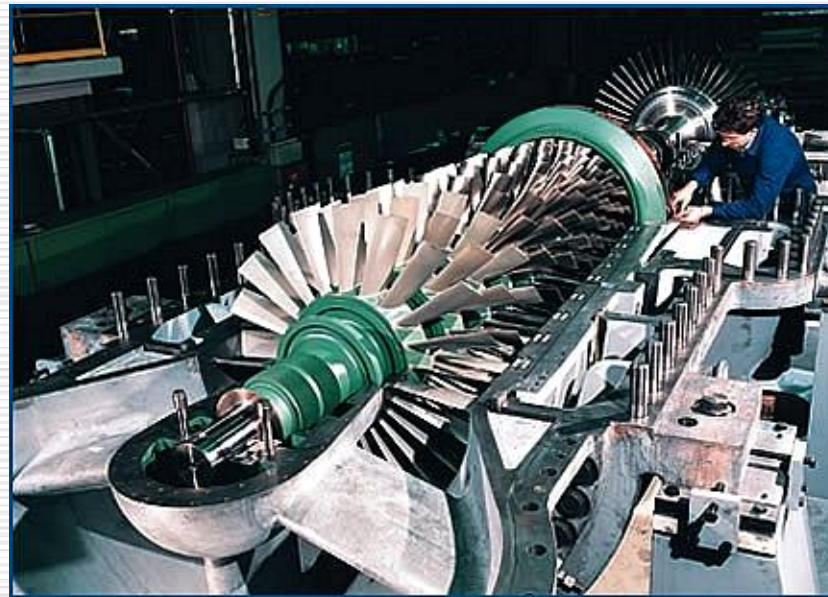


壓縮機簡介

各型式動力式壓縮機簡介-

軸流式, 固定攻角靜葉

(Axial Type, Fixed Stator Vanes)



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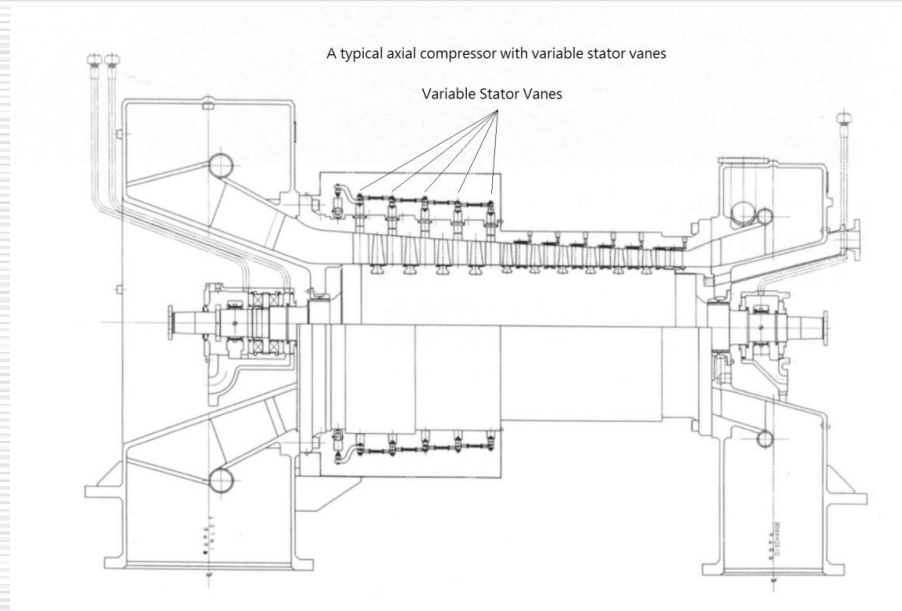


壓縮機簡介

各型式動力式壓縮機簡介-

軸流式, 可變攻角靜葉

(Axial Type, Variable Stator Vanes)

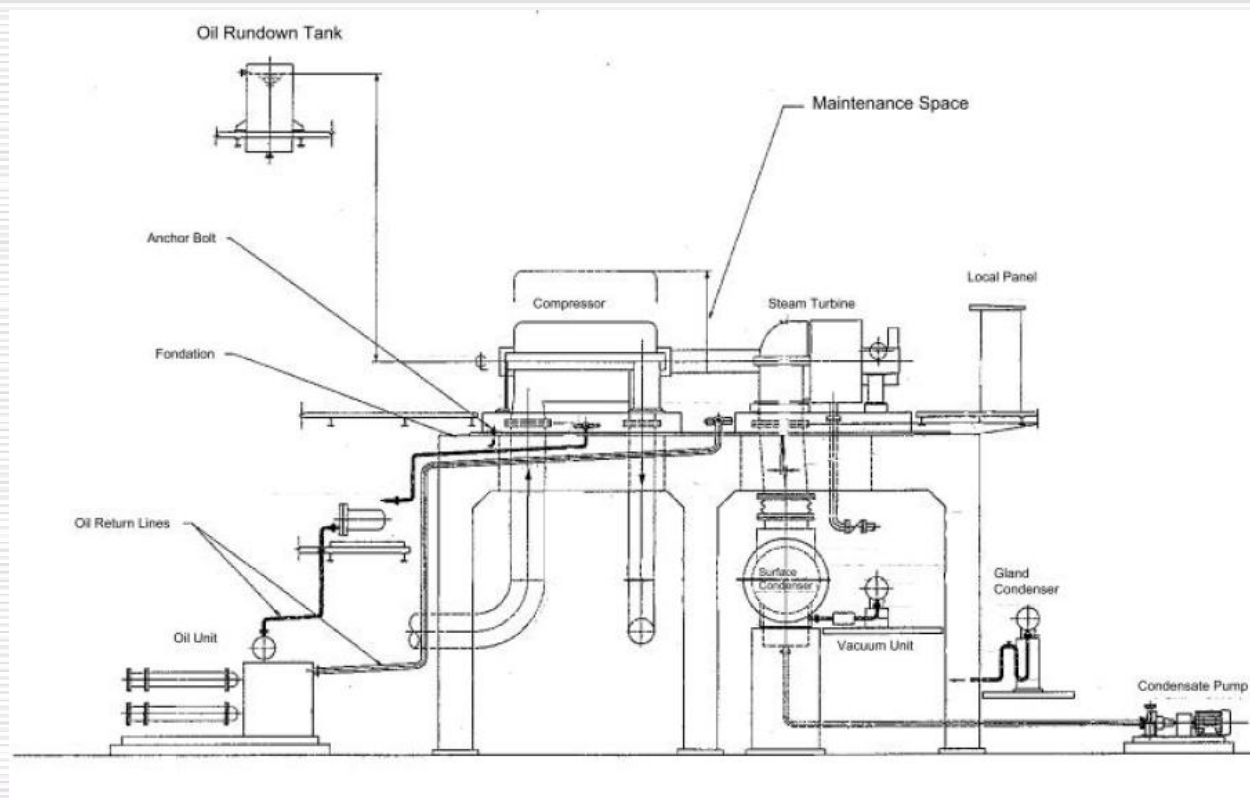


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壓縮機簡介

輔助單元概述



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謝謝！

